

CSC-430 Neural Networks and Deep Learning

Semester: **Fall 2025**

Instructor: Dr. Umer Anwaar

Pre-requisite: Machine Learning or Intro. to AI

Course Objective:

This course provides an in-depth understanding of deep learning concepts, architectures, and applications. It covers topics such as neural networks, convolutional networks, recurrent networks, generative models, and their implementation using popular deep learning frameworks. The course combines theoretical knowledge with practical exercises and projects.

Program Learning Outcomes (PLOs):

- PLO1: Academic Education
- PLO2: Knowledge for Solving Computing Problems
- PLO3: Problem Analysis
- PLO4: Design/Development of Solutions
- PLO5: Modern Tool Usage
- PLO6: Individual and Team Work
- PLO7: Communication
- PLO8: Computing Professionalism and Society
- PLO9: Ethics
- PLO10: Life-long Learning

Course Learning Outcomes (CLOs):

- CLO1: **Understand and explain** the theoretical foundations of deep learning.
- CLO2: **Apply** neural network models using deep learning frameworks.
- CLO3: **Develop/Construct** deep learning techniques to solve real-world problems.
- CLO4: **Analyze** the ethical implications of deploying deep learning models.

Course Outline and Schedule

Week	Topics	Map with CLOs	Map with PLOs
Wk-1	Introduction, Machine Learning Basics	(Refresher), CLO1, CLO2	PLO1
Wk-2	Capacity, overfitting and underfitting, Hyperparameters and validation sets	CLO1, CLO2	PLO1
Wk-3	Deep Feedforward Networks, Regularization	CLO2	PLO2

Wk-4	Convolutional Neural Networks (CNNs) Architectures, Applications	CLO2	PL02
Wk-5	Advanced CNN Architectures, Transfer Learning, Fine-tuning	CLO3	PL03
Wk-6	Dimensionality Reduction, Autoencoders	CLO2	PL02
Wk-7	Generative Models, Variational Autoencoders	CLO3	PL03
Wk-8	Recurrent Neural Networks (RNNs)	CLO3	PL03
Wk-9	Midterm Exam		
Wk-10	Attention and Transformer Architecture	CLO2, CLO3	PL02, PL03
Wk-11	Ethics in AI, Bias and Fairness in Deep Learning	CLO4	PL08
Wk-12	Applications of Deep Learning	CLO3	PL03
Wk-13	Applications of Deep Learning	CLO3	PL03
Wk-14	Applications of Deep Learning	CLO3	PL03
Wk-15	Group Projects on Deep Learning Applications, Supervision and Feedback Sessions	CLO3	PL03
Wk-16	Final Project Presentations, Review of Key Learning Points, Future Directions	CLO3	PL03
Wk-17	Revision (Last week of Classes)		
Wk-18	Final Exam Week (Mon-Fri)		

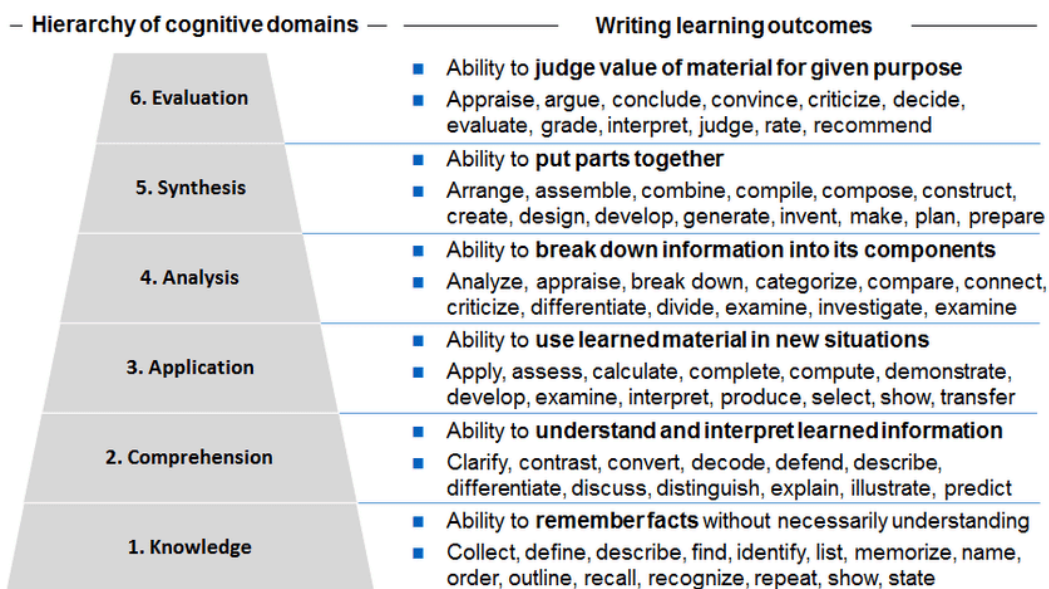
Grading Breakdown and Policy

Assessment Description	Weight (%)	CLOs	PLOs
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Quizzes / Assignments	20%	CLO1, CLO2	PLO1, PLO2
Project (Assigned in the last month)	30%	CLO3, CLO4	PLO4, PLO5, PLO7
Mid Term Exams	20%	CLO1, CLO2, CLO3	PLO2, PLO3, PLO4
Final Exam	30%	CLO1, CLO2, CLO3	PLO2, PLO3, PLO4

The assessment breakdown aligns various assessments with specific Course Learning Outcomes (CLOs) and Program Learning Outcomes (PLOs) to ensure comprehensive coverage of course objectives and competencies. Quizzes and assignments (20%) focus on foundational knowledge (CLO1, CLO2) and basic skills (PLO1, PLO2), while the project (30%) targets higher-order skills (CLO3) and advanced competencies like teamwork and complex problem-solving (PLO4, PLO5, PLO7). Both the midterm (25%) and final exams (25%) cover a broad range of course content (CLO1, CLO2, CLO3) and evaluate intermediate to complex skills (PLO2, PLO3, PLO4), ensuring that all critical aspects of the course and program are assessed.

How to map your CLOs?



References:

1. <https://tips.uark.edu/using-blooms-taxonomy/> (CLOs mapping details)
2. <https://www.hec.gov.pk/english/services/universities/RevisedCurricula/Documents/2016-2017/COMPUTER-SCIENCE.pdf> (PLOs details)